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ORIGINAL PAPER

Antihyperalgesic Effect of Paeniflorin Based on Chronic Constriction Injury in Rats

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Abstract

This study explored paeoniflorin potential to alleviate mechanical pain and cold pain. Behavioral tests were conducted in rats after surgery to evaluate paeoniflorin effect in the pain model, chronic constriction injury of sciatic nerve. After a pharmacological network analysis, it was found that inflammatory mediator regulation of TRP channels was the preferred target for the modulation of chronic neuropathic pain. To verify this result, a series of molecular docking studies with paeoniflorin was conducted. After chronic constriction injuries, rats were sensitized to mechanical hyperalgesia and cold hyperalgesia. The mRNA levels of transient receptor potential cation channels of the subfamily V (vanilloid) 2 (TRPV2) and 3 (TRPV3) and the expression levels of TRPV1 and TRPV4, and those of the subfamilies M (melastatin, TRPM8), A (ankyrin, TRPA1), and phosphorylated p38 mitogen-activated protein kinases (MAPK) increased sharply in the injured animal group. The serum levels of the pro-inflammatory cytokines, such as interleukin 6 (IL6), and tumor necrosis factor (TNF-α) in the injured group were higher. However, paeoniflorin decreased these parameters, and TRPM8 and TRPV1 levels were restored to normal. From immunofluorescence experiments, the levels of TRPA1 and TRPV1 increased dramatically and, after treatments, the levels of TRPA1 and TRPV1 decreased. Therefore, paeoniflorin alleviated chronic neuralgia by decreasing inflammatory factors, suppressing p38 MAPK phosphorylation and reducing the expression of TRP proteins.

Keywords Herbal product · Network pharmacology · Hypersensitivity · Inflammation · Inflammatory mediators · Nociceptor

Introduction

Chronic neuropathic pain was defined as a type of pain that lasted for over 3 months. The prevalence of chronic neuralgia varies from 10 to 11% in adults, affecting millions of individuals throughout the world (Dahlhamer et al. 2018). Chronic neuralgia is a serious type of neuropathic pain that burdens the global economy and public health. Standard analgesics are not usually helpful in relieving neuropathic pain, although the use of opiates is appropriate for clinical use. However, the use of opioids such as morphine, tramadol, and hydrocodone could be expected to be addictive. Moreover, an overdose of opiates could lead to a rise in opiate overdose and mortality, which results in problems with prescribing opioid practices. Thus, it is of prime importance to identify novel nonopioid, nonaddictive drugs for the efficacious treatment of chronic neuropathic pain (Singh et al. 2018). Humankind has used herbal products and phytochemicals to reduce inflammation and pain due to their antihyperalgesic effects which could be use as sources to uncover active plant neuroprotective principles (Singh et al. 2018).

Chronic neuropathic pain was significantly associated with symptoms involving increased reaction and peripheral nerve sensitivity, such as cold hyperalgesia and mechanical hyperalgesia. A growing number of surveys have suggested that inflammatory factors, nerve growth factors, and dorsal root ganglions (DRGs) contributed substantially to the genesis and maintenance of chronic pain (Kinfe et al. 2019;

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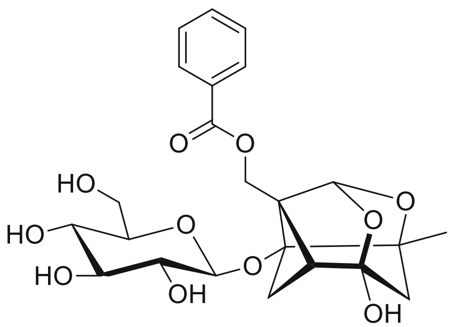
Chapman et al. 2020). Previous research showed that the multicomponent decoction *danggui sini* of seven herbs (Zhang et al. 2019; Pan et al. 2021) can relieve chronic pain. *Tetrapanax papyrifer* (Hook.), K.Koch, Araliaceae, and hederagenine restrained cold hyperalgesia in CCI rats and mechanical hyperalgesia, cold hyperalgesia, and thermal hyperalgesia by reducing the levels of TRPA1, TRPM8, and TRPV1-4 in DRG neurons (Zhang et al. 2020). Paeoniflorin (**1**) is the major effective constituent in the water/ethanol extracts from *Paeonia lactiflora* Pall., Paeoniaceae, which is one ingredient of *danggui sini* decoction (Cheng et al. 2018) that can be used to cure pain and febrile diseases. The paeoniflorin content (more than 1 mg/g) is used to assess the quality of *P. lactiflora* in the People's Republic of China Pharmacopoeia. Paeoniflorin has a neuroprotective effect (Zhang et al. 2021b). Paeoniflorin inhibited Schwann cell injury and apoptosis induced by hydrogen peroxide by attenuating ROS accumulation and inhibiting the phosphorylation of p38MAPK (Zhang et al. 2021b).



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Network pharmacology evaluates the synergistic effects and underlying mechanisms of multicomponent and multitarget agents using network analysis, which is an appropriate way to measure the effectiveness and demonstrate the functional mechanisms of novel bioactive compounds. In this study, we predicted some signaling pathways activated by paeoniflorin (**1**) in the treatment of chronic neuralgia *via* network pharmacology. Thus, it was considered that inflammatory mediator regulation of transient receptor potential (TRP) cation channels was closely related to this process. By experimental verification, it was proved that paeoniflorin alleviated chronic neuropathic pain by decreasing inflammatory factors, suppressing p38 MAPK phosphorylation, and reducing the expression of TRP proteins.

Materials and Methods

Acquiring Protein Targets

By inserting the keywords chronic neuropathic pain and chronic neuralgia into GeneCards (https://www.genecards.org), some chronic neuropathic pain-related genes were identified.

Predicting Paeoniflorin Protein Targets

SMILES of paeoniflorin (**1**) was obtained from PubChem (https://pubchem.ncbi.nlm.nih.gov/). We logged on to Swiss Target Prediction (http://www.swisstargetprediction.ch/) and entered SMILES with species limited to *Homo sapiens*. The obtained protein targets and corresponding UniProt ID were imported into Excel. In addition, the common targets of paeoniflorin and pain were acquired.

Pathway Analysis

DAVID (https://david.ncifcrf.gov/) provides large-scale functional annotations. The targets were imported into DAVID with species limited to *Homo sapiens*. GO analysis and KEGG pathway analysis were conducted.

Molecular Docking

To further validate the robust association between paeoniflorin (**1**) and TRPs, molecular docking was performed. First, the 2D structure of paeoniflorin from PubChem was acquired, and we acquired 3D protein structures of TRPs from the RCSB PDB database. The PyMOL software was used to extract the original ligand conformation of target protein. After preparation of the ligand and receptor molecule, we conducted molecular docking with AutoDock vina. After adding the polar hydrogen and Gasteiger charges to the presented receptors and ligands, semiflexible docking was employed. The binding energy (≤ 5.5) was used to evaluate the molecular docking results. Finally, we applied the PyMOL software to analyze and visualize the docking results of paeoniflorin and proteins.

Animals

Thirty-two male Sprague-Dawley (SD) rats, weighing nearly 200–230 g, aged 7–8 weeks, were acquired from the Experimental Center of Beijing Huafu Kang Co., Ltd. Five rats were individually housed and maintained at 20–22 °C room temperature, 40–60% humidity, and a 12-h light/dark room in the Animal Experimental Center of Medical College of Jinan University. All rats were allowed to adapt to the new environment for 10 days before starting the experiments.

Neuropathic Pain model

Pentobarbital sodium (3%, 40 mg/kg) was injected into rats intraperitoneally. Rats were fixed, and we exposed the right sciatic nerve. Under a microscope, using 4.0 sutures, the right sciatic nerve was tied and then conducted 4 times with intervals of approximately 1 mm. The ligation site is located before the nerve bifurcates. In this way, we can observe a little twitch in the right operated hind limb. However, rats in the sham group did not undergo nerve ligation. Following the incision, gentamicin (10 mg/ml, *i.m.*) was given in the right bicep femoris (Choi et al. 1994; Zhang et al. 2020)

Treatment Solutions

Paeoniflorin (S31585, purity 97%) was acquired from Guangzhou Jiayan Co., Ltd. (Guangzhou, China). Pregabalin (Lyrica) was acquired from Pfizer Manufacturing Deutschland (Freiburg, Germany). Thirty-two male SD rats were randomly divided into 4 groups (8 rats in each group), consisting of the sham operation group (SG), the CCI group (CG), the pregabalin group (PG), and the paeoniflorin group (PFG). These rats received oral treatment for 21 days. As for treatments, rats in SG and CG were given saline (0.9%, 6 ml/kg/bid). Rats in PG were gavaged with pregabalin (10 mg/kg/bid), and rats in PFG were gavaged with paeoniflorin (10 mg/kg/bid) (Ma et al. 2016). The administration began on the first day after surgery and lasted for 21 days.

Behavior Tests

As for cold hyperalgesia, 100 μl acetone was applied onto the plantar surface of the right paw. Then, the total amount of time lifting or clutching the right hind paw was noted during 120 sec. The total amount of time measures the sensitivity to cold pain. Von Frey test was conducted to evaluate mechanical hyperalgesia (50% MWT, mechanical withdrawal threshold). Briefly, rats were placed into a test chamber with a framed metal mesh floor. After 3–5 min habituation, the von Frey monofilaments were applied through the metal floor and thresholds were measured using the updown paradigm. The smallest force used to make rats respond (clear roaring, paw withdrawal, shaking, or licking) was recorded. We calculate the pain threshold as follows: 50% g threshold = $10^{[Xf + k\delta]}/10{,}000$. We test for hyperalgesia on the right paw in the 1st, 4th, 7th, 14th, and 21st days after surgery.

Hematoxylin-Eosin Staining

Sciatic nerve ligation was fixed in 4% paraformaldehyde for 1 day, and then, the paraffin-embedded specimens, specifically the ligation part, were sliced at a thickness of 3 μm for HE staining. Sections were deparaffinized with xylene and incultured with 100, 90, 80, and 70% EtOH. In addition, we rinsed these in PBS for 5 min. After staining, two pathologists who were blinded to experiments observed the photos and assessed tissue damage.

ELISA Test

Rat serum was used for the ELISA Test. First, we anaesthetized rats with pentobarbital sodium and then extracted about 8 ml of blood through the abdominal aorta for every rat. The blood collection was centrifuged at 4032×*g* for 15 min at 4 °C. We obtained 800 μl of the supernatant, which was used to detect levels of IL6, and TNF-α. Experimental operations were conducted by the instructions. We used a microplate reader to analyze the optical density value.

Quantitative Real-Time PCR

Total RNA was extracted from right DRGs (L4–L6, 3 rats) by employing the RNAiso plus. The micronucleic acid spectrophotometer to measure RNA concentration was used. Then, RNA was synthesized into cDNA following the instructions of the T-PCR Kit. Lastly, qPCR was carried out on Applied Biosystems 7900 real-time PCR systems with SYBR Green Premix qPCR and the primers were as follows (Zhang et al. 2021a).

Gene	Forward primer (5′ ≥ 3′)	Reverse primer (5′ ≥ 3′)
β Actin	CCTAGACTTCGAGC AAGAGA	GGAAGGAAGGCTGG AAGA
TRPV2	AGGCGAAGTTGAAG AAGAAT	CGAAAGTTTACTGA GTGGTGT
TRPV3	CCGAGGACGGTAGT AAGA	GTGGACACGACGAC AGAT

Western Blots

The expressions of TRPV1, TRPV4, TRPA1, TRPM8, p38MAPK, and p-p38MAPK on DRGs (L4–L6, 4 rats) were analyzed by Western blot. Right DRGs (L4–L6) were removed carefully as previously described (Martinez-Rojas et al. 2014). Then, DRGs were broken into fragments and then were homogenized in RIPA for 20 min at 4 °C. Homogenates were centrifuged at 12,074×*g* at 4 °C. We acquired the protein concentrations via the BCA assay. The amount of protein load was 30 μg per group. After electrophoresis, transmembrane, and blocking, membranes were incubated with primary antibody (1:1000) overnight at 4 °C. The next day, membranes were incubated with secondary antibody (1:5000) for 1 h at room temperature. Last, membranes were detected via the automatic gel imaging system. Image Lab software was used to measure protein expression.

Immunohistochemistry

Right DRGs (4 rats, the same as those in Western blots) were fixed in 4% paraformaldehyde for 24 h. Then, they were transferred into a 20% (w/v) sucrose solution. DRGs were cut at 9-μm sections. These sections from dorsal root ganglion (DRG) neuron were homogenized in sodium citrate antigen repair solution (1:1000 attenuation, pH 6). Primary antibodies were diluted in 1:200. TRPV1 and TRPA1 were used as primary antibodies. Then, the sections were incubated with the corresponding second antibody (1:500). Stained sections were reviewed under a fluorescence microscope (Leica). TRPV1 or TRPA1-expressing positive cells indicated that TRPV1 or TRPA1 were expressed near the nucleus, and the number of these cells was calculated.

Statistical Analyses

Statistical analysis was conducted with GraphPad Prism 8, Origin 2021, and SPSS13.0 software to generate all graphs. The data of the behavior tests were analyzed by repeated measure ANOVA. The other data were analyzed by one-way ANOVA. Tukey's multiple comparisons test was the *post hoc* test after ANOVA. $P < 0.05$ was considered to indicate a statistically significant difference.

Results and Discussion

Target Predictions

First, SMILES of paeoniflorin was obtained from the PubChem, which was CC12CC3(C4CC1(C4(C(O2)O3)COC(=O)C5=CC=CC=C5)OC6C(C(C(C(O6)CO)O)O)O)O. After inputing the 2D structure of **1** into Swiss Target Prediction, 100 paeoniflorin therapeutic target were predicted (Table S1). Meanwhile, 1167 targets of chronic neuropathic pain from GeneCards were identified. Taking the intersection of the two parts, 38 common targets were acquired (Table S2).

Paeoniflorin Treatment

The 1167 identified targets of chronic neuropathic pain and 38 common targets were processed by DAVID. Then, we obtained KEGG enrichment analysis, and chose the first 20 pathways, drawing bubble charts via Sanger assistant SangerBox (http://sangerbox.com/). Figures S1A and 1B showed some mechanisms underlying chronic neuropathic pain and paeoniflorin in the treatment of chronic neuralgia, respectively. In these two images, the inflammatory mediator regulation of TRP channels ranked first, simultaneously, indicating that the regulation of this signaling pathway is closely related to the occurrence and relief of chronic neuralgia. Therefore, it is quite reasonable to choose this target for molecular docking and experimental validation with paeoniflorin (**1**).

GO Analysis of Target Genes

Gene ontology (GO) analysis found 117 biological processes, 22 cellular components, and 49 classes of molecular functions which were involved in the mechanism underlying the analgesic effect. The first 10 classifications and the targets were drawn in Figure S1C–E via origin 2021.

Alluvial Diagrams for Paeoniflorin-Target-Pathways

Alluvial Diagram (Figure S1F) via Origin 2021 Based on the chart, we clearly saw that paeoniflorin alleviated neuropathic pain through specific targets related to the regulation of signaling pathways, which showed the interrelationship among paeoniflorin, therapeutic targets, and signaling pathways.

Molecular Docking

To further verify the binding capacity between paeoniflorin and TRPs, molecular docking analysis was performed. It is generally accepted that the lower the binding energy indicates the greater the possibility of the effect. Four genes (TRPA1, TRPM8, TRPV1, and TRPV4) and paeoniflorin were employed in molecular docking analysis. We acquired the PDB codes 6V9 V (Zhao et al. 2020), 6O72 (Diver et al. 2019), 7LP9 (Kwon et al. 2021), and 4DX1 (Inada et al. 2012) from RCSB and these can be referred from the previously published study. The binding energy − 7.3, − 9.2, − 8.0, and − 5.9 kJ/mol, respectively, were found for paeoniflorin-TRP interaction (Fig. 1A–D). The complex paeoniflorin-TRP was stabilized by hydrogen bonds with selected residues, which are shown in Fig. 1A–D, confirming the reliability of network pharmacology prediction. Table 1 summarizes the binding site and hydrogen bond lengths.

Behavioral Results

Figure 2A and B demonstrate that rats in the CCI group exhibited mechanical hyperalgesia and cold hyperalgesia 4 days after the operation. Rats in the paeoniflorin group were more insensitive to mechanical and cold stimulation (7, 14, and 21 days than those in the CCI group. Although paeoniflorin relieved mechanical pain and cold pain, they were not as normal as those in the sham group (7, 14, and 21 days). Except for mechanical pain at the 21st day, there was no difference between the paeoniflorin group and the pregabalin group. Pregabalin relieved mechanical pain better.

HE Staining

In the 21st day, the structure of sciatic nerve in the sham operation group can be spotted. In these staining (Fig. 3A; $n = 8$), Schwann cells and inflammatory cell infiltration were

Table 1 Interactions between paeoniflorin (**1**) and transient receptor potential channels by molecular docking

	Binding energy (kJ/mol)	Binding site	Hydrogen bond length (Å)
Paeoniflorin-TRPA1	− 7.3	THR-684, LYS-661	2.2, 2.1
Paeoniflorin-TRPM8	− 9.2	ARG-832, GLN-776	2.0, 2.0, 1.5
Paeoniflorin-TRPV1	− 8.0	ASN-437, ARG-491, TYR-487	2.3, 1.9, 2.4, 2.2, 2.6, 2.4
Paeoniflorin-TRPV4	− 5.9	ASP-396, GLU-393	2.1, 2.0

hardly observed. The inflammatory cell infiltration was observed effortlessly, and the proliferation of a large number of Schwann cell was spotted smoothly (Fig. 3B). In addition, the normal structure of the sciatic nerve was damaged in the CCI group (Fig. 3B). However, after treatments, a decline of inflammatory cells and the Schwann cells was spotted, but organizational structure and morphology of the sciatic nerve were not restored to normal (Fig. 3C, D).

Inflammatory Factors

The levels of the pro-inflammatory cytokine IL6 and TNF-α in serum were measured by ELISA. The expressions of IL6 and TNF-α raised in the CCI group sharply, and pregabalin and paeoniflorin reversed these trends (Fig. 4A and B, $p < 0.05$, $n = 8$), indicating that the antihyperalgesic effects of paeoniflorin and pregabalin are associated with reduced

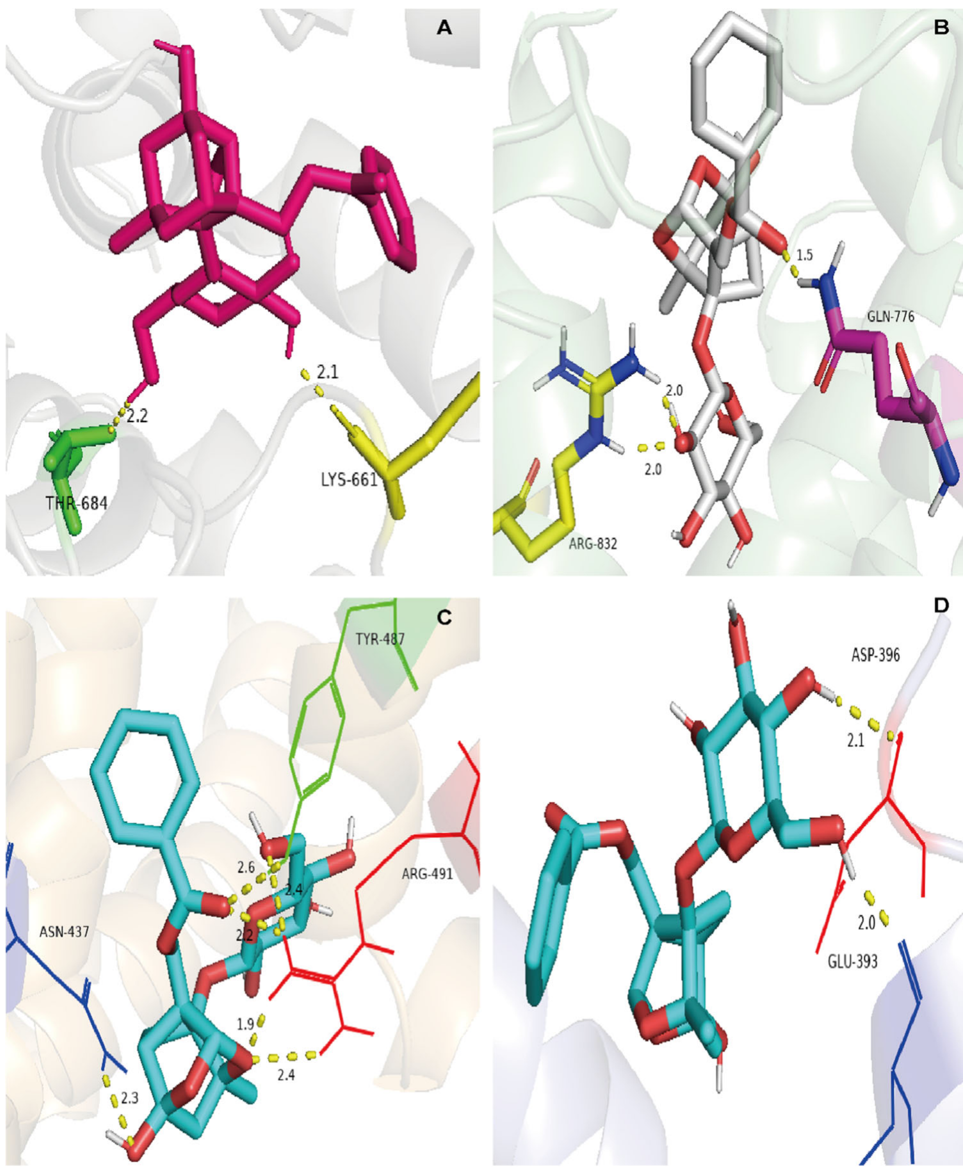


Fig. 1 Interactions between paeoniflorin (**1**) and TRPA1 (**A**), TRPM8 (**B**), TRPV1 (**C**), and TRPV4 (**D**)

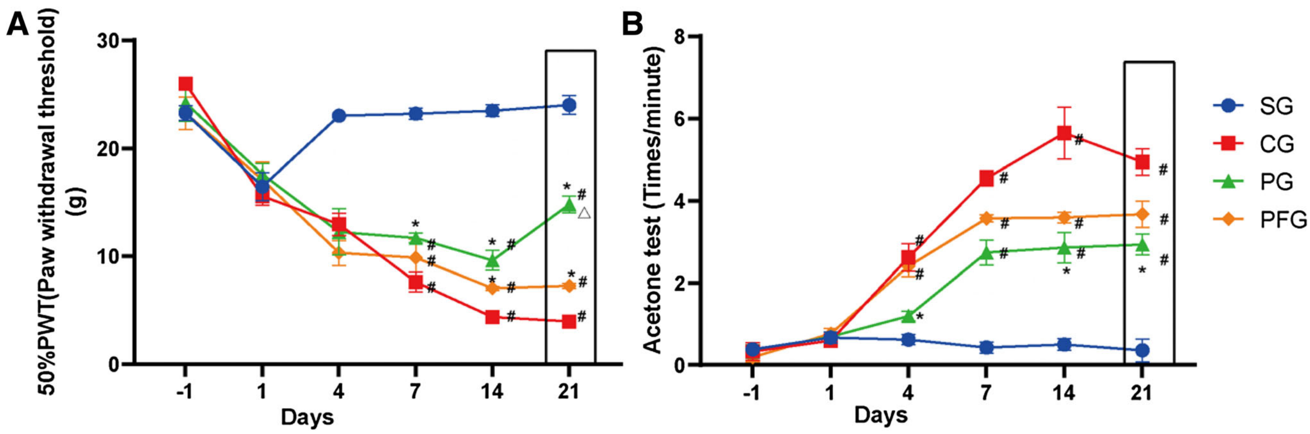


Fig. 2 Antihyperalgesic effect of paeoniflorin to chronic constriction injury of the sciatic nerve on pain. **A** The tested rats in the treated group were insensitive to mechanical pain (7, 14, and 21 days; $p < 0.05$). **B** The tested animals were insensitive to cold stimulation compared to those in the CCI group (7, 14, and 21 days, $p < 0.05$). #Comparison of the sham operation group and others. *Comparison of the CCI group and others. △Comparison of paeoniflorin group and pregabalin group. The maxima effects that paeoniflorin produced are marked in the rectangle

neuralgia and inflammation. In the pregabalin group, IL6 levels declined to nearly normal levels. Paeoniflorin was not able to restore the expressions of IL6 and TNF-α to baseline levels (Fig. 4A and B, $p < 0.05$), which may explain why pregabalin is better than paeoniflorin in treating mechanical pain.

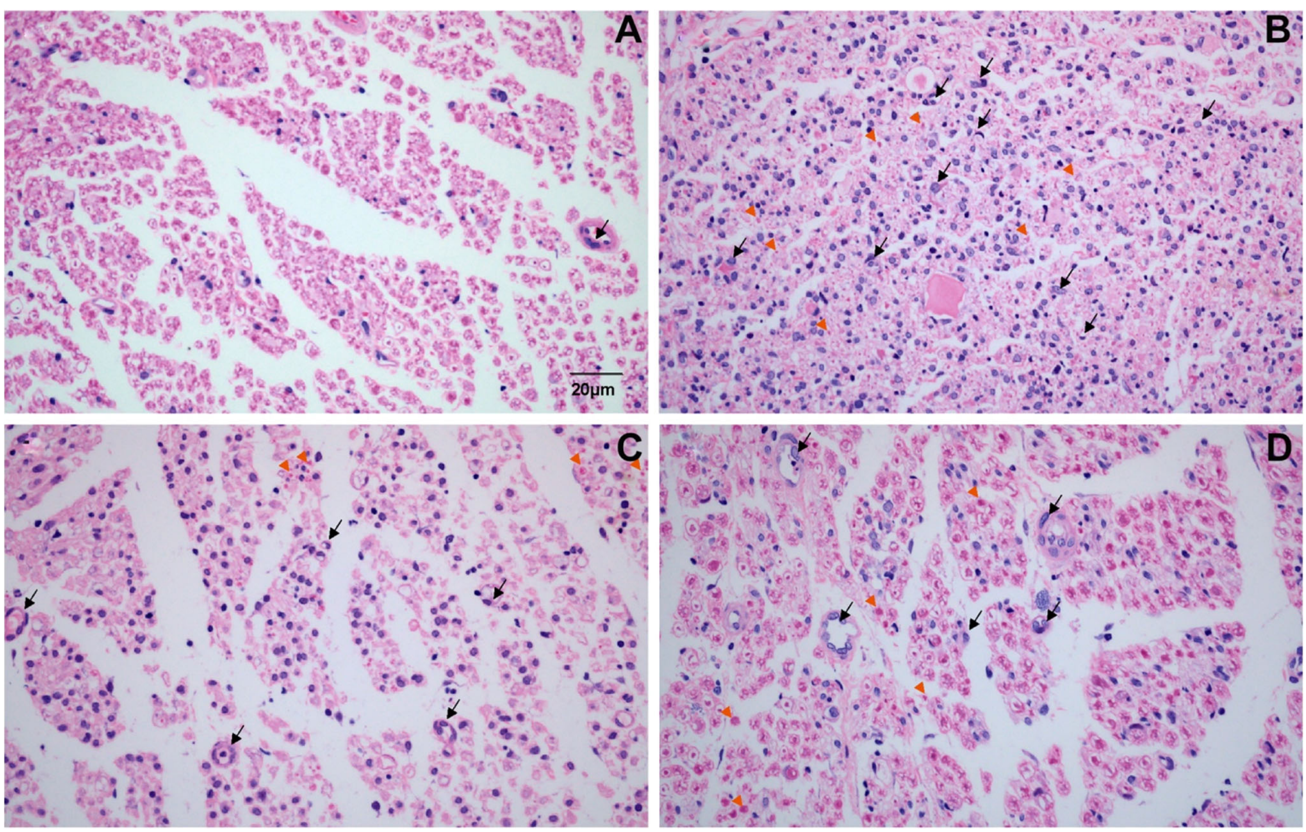


Fig. 3 The structure of sciatic nerve in sham operation group can be identified. The structure of sciatic nerve in sham operation group can be identified (**A**). We hardly observed Schwann cells, as well as inflammatory cell infiltration (**B**). The inflammatory cell infiltration and many Schwann cell proliferations were easily observed. In addition, the normal structure of the sciatic nerve was damaged in the CCI group (**B**), and the organizational structure and morphology of the sciatic nerve were not restored to normal (**C** and **D**). Black marked—Schwann cells and orange marks—inflammatory cells

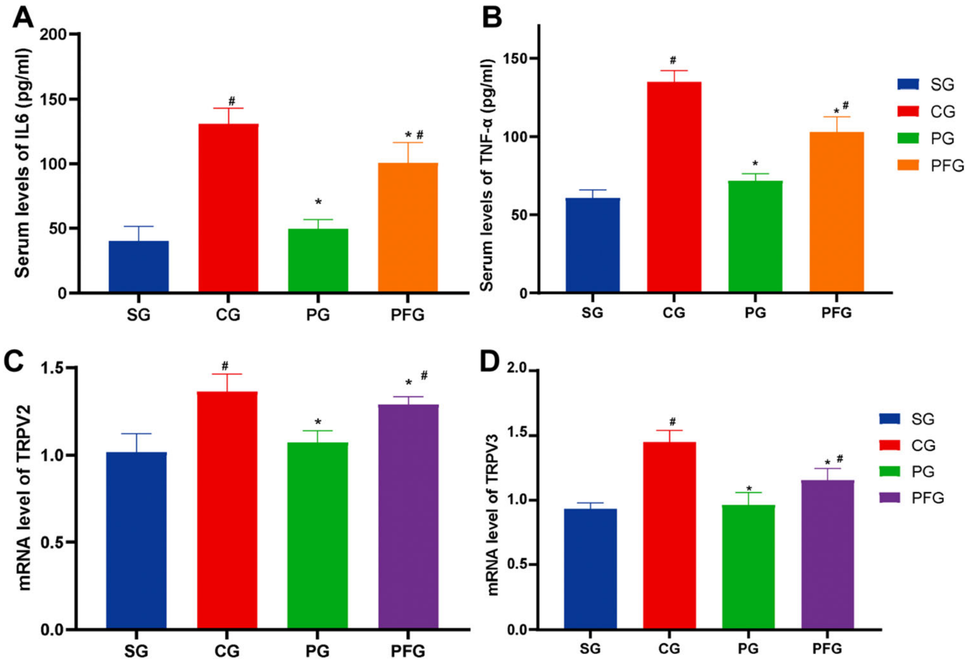


Fig. 4 Pregabalin and paeoniflorin can lower the expressions of IL6, and TNF-α in serum (**A** and **B**, $p < 0.05$). After CCI surgery, mRNA levels of TRPV2 and TRPV3 increased moderately (**C** and **D**), and treatments can relieve these. #Comparison of the sham operation group and others. *Comparison of the CCI group and others

TRPV2 and TRPV3 Levels

The mRNA expressions of TRPV2 and TRPV3 in dorsal root ganglion (DRG) in the CCI group increased sharply in the 21st day following operations (Fig. 4C and D, $p < 0.05$, n = 3). Pregabalin deteriorated the mRNA expressions of TRPV2 and TRPV3 to nearly normal, while paeoniflorin was not able to restore these to normal.

Expressions of Transient Receptor Potential Channels

In the CCI group, the levels of TRPM8, TRPA1, TRPV1, TRPV4, and p-p38MAPK increased significantly after 21 days following CCI operations ($p < 0.05$). Paeoniflorin and pregabalin reduced the expressions of TRPM8, TRPA1, TRPV1, TRPV4, and p-p38MAPK (Fig. 5, $p < 0.05$, n = 3). Pregabalin declined all these to the normal except TRPV4 while paeoniflorin can only downregulate the level of TRPM8 to normal. These results explained why pregabalin is better than paeoniflorin in relieving mechanical pain.

Immunohistochemistry of DRG Tissues

Immunofluorescence results show the expression of TRPA1 and TRPV1 in the DRGs (n = 3, L4–L6) altered in varying degrees in different groups (Fig. 6). After surgery, compared with the sham operation group, the levels of TRPA1 (A) and TRPV1 (B) increased sharply. With 21 days of treatment, the positive cells of TRPV1 and TRPA1 in other groups were much lower than those in CCI model group ($p < 0.05$). These results are consistent with the results of Western blot.

In this study, it was revealed that rats in the CCI group were more sensitive to mechanical, hot, and cold stimuli than those in the sham operation group. In addition, paeoniflorin can prevent the development of chronic neuralgia and relieve chronic neuropathic pain. The original and major findings are as follows: (1) the pathway named inflammatory mediator regulation of TRP channels were close to chronic neuropathic pain; (2) paeoniflorin inhibited serum inflammation and reduced the inflammatory cells in sciatic nerve; (3) paeoniflorin inhibited neuralgia by inhibiting the inflammatory mediator regulation of TRP channels pathway; (4) the expressions of TRPA1 and TRPV1 in DRGs were higher after sciatic nerve injury.

TRPA1 and TRPM8 could be triggered by cold or cooling agents (Peier et al. 2002; Pan et al. 2018), while TRPA1 was also a mechanical sensitivity of receptor, which was related to the pathogenesis of mechanical hyperalgesia (Nassini et al. 2014). TRPM8 played considerable roles in the sensitization of nociceptive transduction (Basso and Altier 2017). Luo et al. (2019) alleged that (−)-menthol activated TRPA1 and TRPM8 channels to increase the excitatory synaptic transmission. TRPV1 and TPRV4 are triggered by distinct heat temperatures and mechanical stimuli and are tightly contacted with mechanical hyperalgesia and thermal hyperalgesia (Qu et al. 2016; Gouin et al. 2017). Chronic pain can be relieved by blocking the TRP pathway as previously described (Chia et al. 2020). In this study, we found that the levels of TRPV1, TRPV4, TRPM8, TRPA1, and phosphorylated p38MAPK increased clearly after CCI operation, and treatments lowered these. Yin et al. (2016) considered that paeoniflorin exerted antihyperalgesic effects and might be of potential use

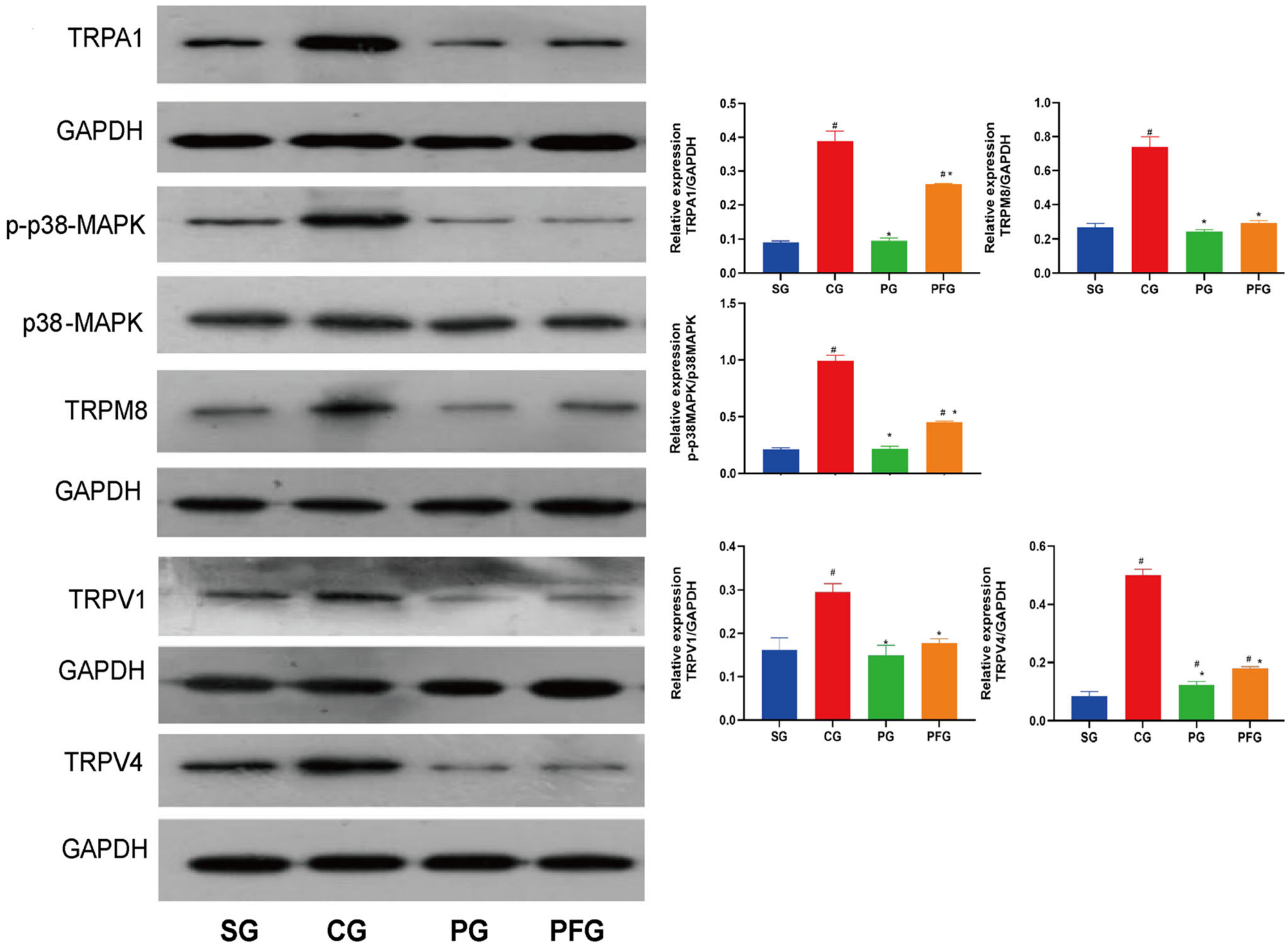


Fig. 5 Expressions of TRPA1, TRPM8, TRPV1, TRPV4 and phosphorylated p38 MAPK. The levels of TRPM8, TRPA1, TRPV1, TRPV4, and p-p38MAPK increased significantly 21 days following CCI operation ($p < 0.05$). #The contrast of the sham operation group and others. *The comparison of the CCI group and others

in the treatment of chronic neuropathic pain. Therefore, paeoniflorin relieved pain because its significant blocking action of TRP channels, including TRPA1, TRPM8, and TRPV1-4, are closely related to the occurrence of chronic pain (Scheme 1).

Inflammation is the result of pain, which is activated by several pathways triggering acute or chronic pain. Some studies have shown that paeoniflorin and some compounds with similar structures can also treat pain. Paeoniflorin and albiflorin can attenuate neuropathic pain via the MAPK pathway in rats with chronic constriction injury (Zhou et al. 2016). Wang et al. (2014) identified that paeoniflorin and albiflorin could inhibit the expression of IL-6 and TNF-α. Jian-Yu Zhou et al. (2016) revealed that both compounds inhibited the activation of p38MAPK to reduce pain. Albiflorin reduced the activation of calmodulin-dependent protein kinase II and c-Jun *N*-terminal kinase in the hypothalamus (Zhang et al. 2016). Paeonol ameliorated CFA-induced inflammatory pain by inhibiting the HMGB1/TLR4/NF-κB p65 pathway (Qiu et al. 2021). Zarpelon et al. (2013) found that inflammatory mediators sensitized TRPV1 nociceptors, enhanced pain sensation, and led to hyperalgesia. Chen and Hackos (2015) indicated that inflammatory environments or tissue injury sites could activate TRPA1, and our study was consistent with this finding.

In this study, paeoniflorin inhibited the progression of chronic neuralgia and stopped the pain, including mechanical hyperalgesia and cold hyperalgesia, from worsening. In addition, it reduced the level of inflammatory factors and promoted injured sciatic nerve repair. We concluded that paeoniflorin cured chronic neuralgia by reducing inflammatory factors inhibiting the p38 MAPK pathway and decreasing the expression of TRP proteins via network pharmacology prediction and experimental validation.

Conclusion

According to network pharmacology, inflammatory mediator regulation of TRP channels was closely related to chronic

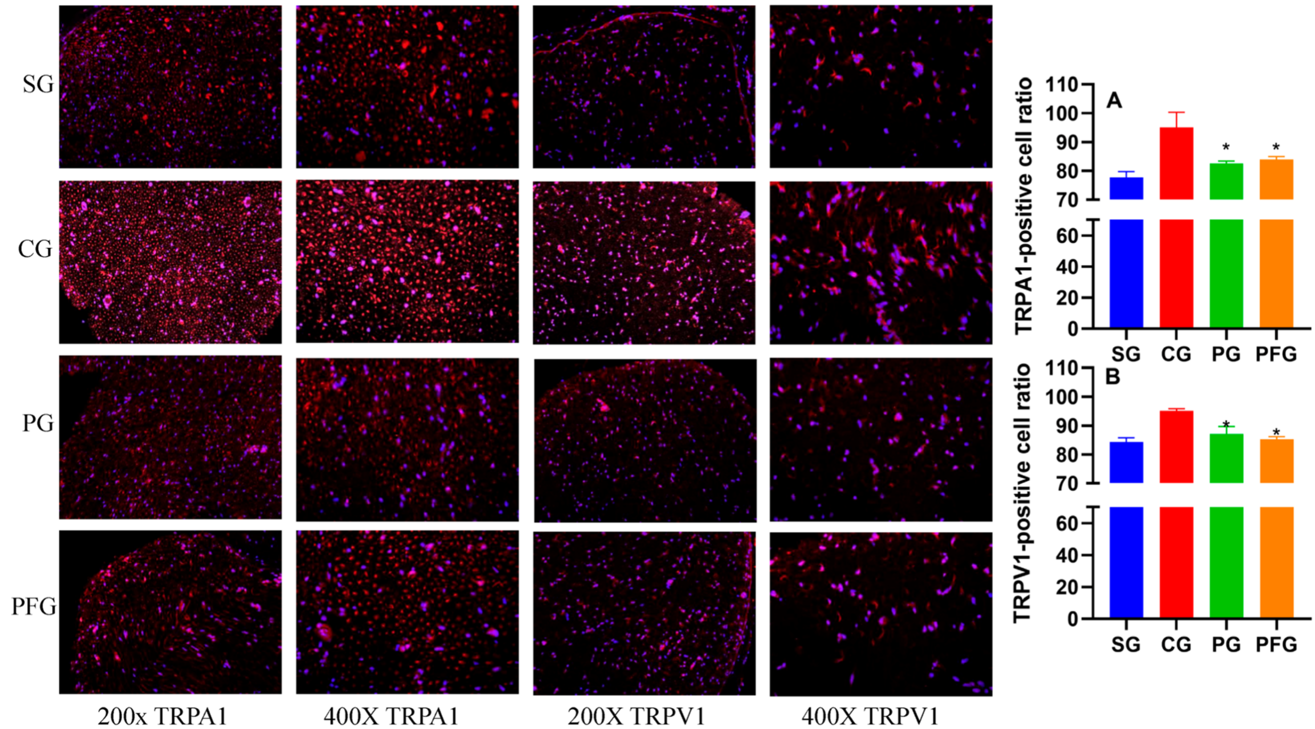
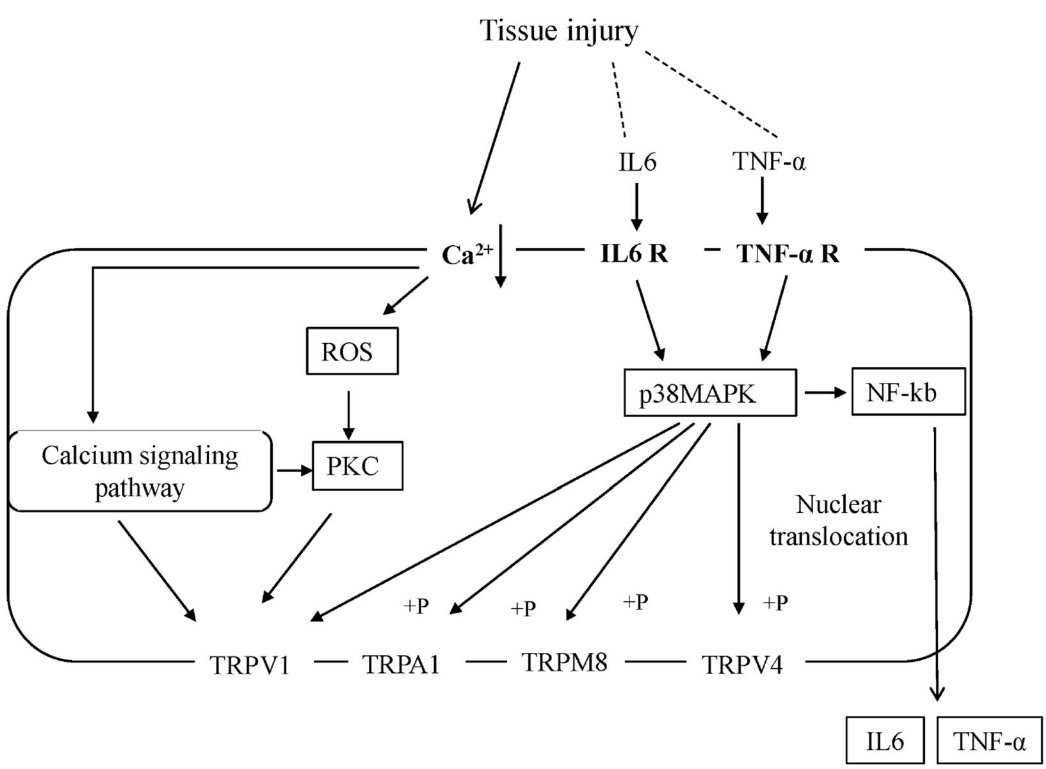


Fig. 6 Immunofluorescence pictures of TRPA1 and TRPV1 compared to those in sham operation group; the levels of TRPA1 and TRPV1 increased sharply. After treatments, the levels of TRPA1 and TRPV1 decreased ($p < 0.05$). *Comparison of the CCI group and other groups

neuropathic pain. In the experimental phase, CCI rats revealed mechanical hyperalgesia, cold hyperalgesia and nerve structure was damaged. Increased levels of IL6 and TNF-α in serum, and increased expressions of TRPV1, TRPV4, TRPA1, TRPM8, and the phosphorylation of p38MAPK were observed. After receiving treatments, rats increased their



Scheme 1 Signaling pathway diagram during chronic pain

tolerance to pain and inflammatory factors and TRP protein levels declined. In conclusion, paeoniflorin treated chronic neuralgia by reducing inflammation factors inhibiting p38 MAPK pathway and decreasing the expression of TRP proteins.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s43450-022-00251-z.

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Author Contribution DZ, BJ, JS, and GZ made considerable contributions to the experimental design, data analysis, and experimental procedures. XL, HS, ZC, SC, YZ, and YP helped us with experimental procedure and English writing. Thank YL (Yixuan Li) for her valuable comments on English writing. GZ and JS are the corresponding authors.

Declarations

Ethical Standards The studies were compliant with Jinan University Ethics Committee standards (approval code: CJU2021015A, October 8, 2018).

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